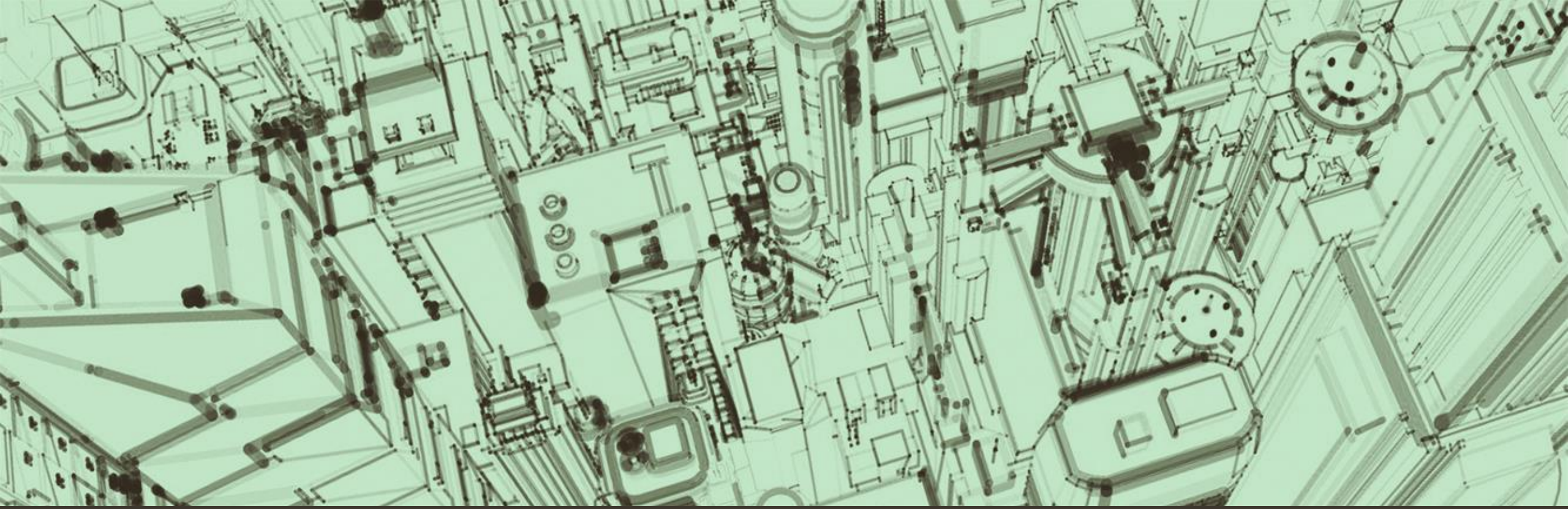


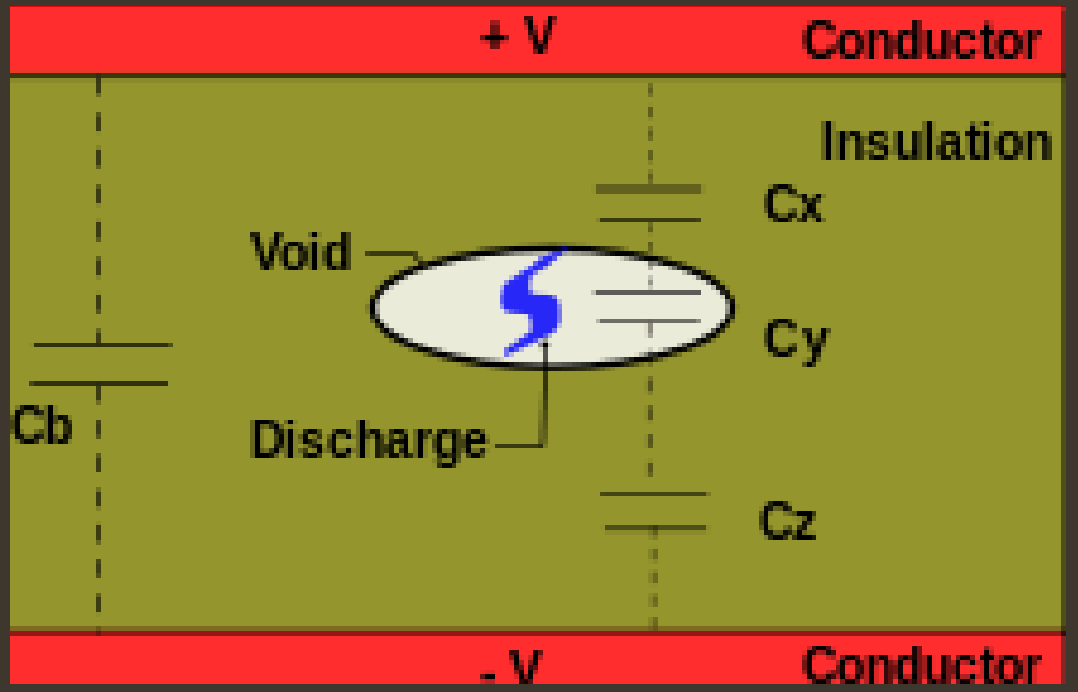
PARTIAL DISCHARGE TEST PEHLA LABORATORY

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What is a partial discharge?

- Localized electrical discharges inside the insulating system of electrical equipment are referred to as partial discharges.
- These discharges often have a tiny size and take place in insulation problems, gaps, or spaces that are filled with air.



Partial Discharge Testing's Importance

- An important diagnostic tool for determining the state of insulation systems in different types of high-voltage equipment is partial discharge testing.
  It assists in finding potential insulation weaknesses, analyzing insulation effectiveness, and ensuring the equipment's stability and security.

WHAT IS THE PURPOSE OF PARTIAL DISCHARGE TEST?

1. Evaluating Insulation Quality: This test analyzes the quality and safety of the insulation systems in electrical equipment, locating any weaknesses or defects that may lead to device failure.

2. Early Insulation Issue Detection: The test enables early detection of possible issues before they develop by recognizing partial discharges, which indicate insulator degradation.

3. Catastrophic failures can be avoided by early detection of insulation breakdown, which protects employees and equipment from dangerous activity such as short circuits and electrical fires.

4. Maintaining Equipment Reliability: By providing the integrity of its insulation system, the test makes sure that electrical equipment, such as transformers and motors, maintains reliable performance.

5. Improving Maintenance Strategies: Consistent partial discharge testing offers useful information for maintenance planning, allowing proactive actions like preventative maintenance and prompt repairs or replacements.

6. Attention to Standards: A number of industrial norms and rules call for partial discharge testing of particular equipment. These tests are performed to ensure standards are followed and to improve the overall quality and security of the system.

THE EQUIPMENT USED FOR TESTING

TEST OBJECTS: Transformators, cables, switchgear, motors, and other electrical devices or components are examples of test objects.

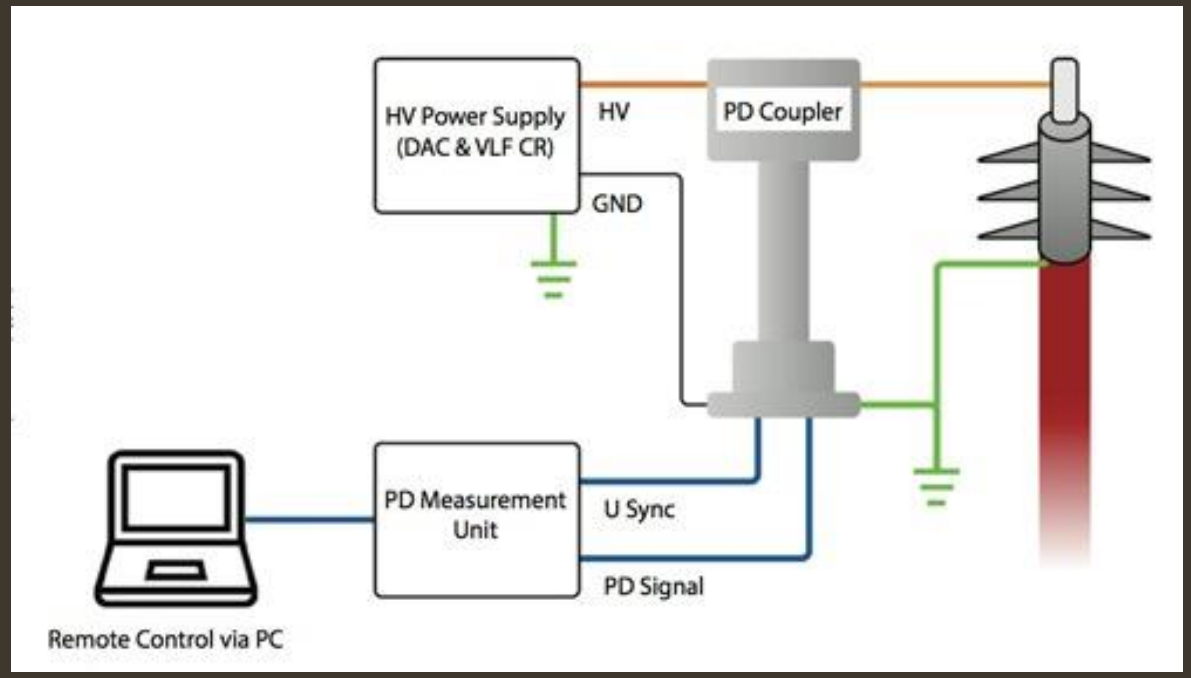
HIGH VOLTAGE POWER SOURCE: An electrical source able to deliver the necessary voltage for the test. In accordance with the particular test requirements, it may be an AC or DC power supply.

COUPLING CAPACITOR: A capacitor that links the test object to the high voltage power source. It guarantees that the test object's insulation system receives a high voltage application.

PARTIAL DISCHARGES DETECTORS: Sensors are specialized devices created to record and evaluate the signals produced by partial discharges during the test. These detectors, which may pick up electromagnetic or auditory signals linked to partial discharges, can be capacitive, inductive, or optical sensors.

5. Data Collection System: A tool for collecting and analyzing partial discharge signals from sensors or detectors. Oscilloscopes, data recorders, and specialized partial discharge monitoring devices are frequently included in it.

6. Grounding and Safety Equipment: Important security measures are used for the test, such as grounding wires, security barriers, and personal protective equipment (PPE), to guarantee the safety of workers and equipment



Test steps

1. . Test Preparation:

- Ensure that the test object is installed correctly and grounded.
- Check that the high voltage power supply, partial discharge detector/sensor, and data collecting system are installed correctly.
- Carry out the appropriate calibrations and accuracy tests.

2. Test Setup:

- Use a coupling capacitor to attach the test item to the high voltage power source.
- Set the required test voltage level in accordance with the requirements or standards for the equipment.
- Implement safety precautions including grounding and personal safety gear.

3. Initial Conditioning:

- To condition the test object and clear any particles or moisture from the surface, use a low voltage.
- As the voltage rises gradually to the specified test level, the insulation will stabilize.

Test steps

4. Partial Discharge Measurement:

- Activate the detector or sensor for partial discharges to record signals.
- Use the data collection system to record the signals, either regularly or periodically.
- Continue to keep a watch on and record signals for a predetermined amount of time or until enough information has been collected.

5. Data Interpretation and Analysis:

- Examine recorded signals to identify factors including the size, frequency, phase, and location of the discharge.
- Evaluate the intensity and features of partial discharges by comparing measured values against thresholds or standards.
- Analyze the data to spot any insulation problems or other test items red flags.

6. Test Results:

- Determine the state of the insulation system and the necessary steps based on the analysis.
- Create a test report that summarizes the setup, outcomes, and any maintenance advice.

1- Magnitude of the discharge:

- Peak Magnitude: Determining the discharge signal's greatest amplitude.
- RMS Magnitude: Calculating the discharge signal's root mean square value.

2. Discharge Frequency: It measures the frequency of occurrence by counting the number of discharges per unit of time.

3. Parameters for Discharge Pulse:

- Rise Time: Determining the period of time it takes for the discharge pulse to increase in amplitude from a certain percentage.
- Duration: Calculating the duration of the discharge pulse above a predetermined threshold.
- Fall Time: Determining the amount of time it takes for the discharge pulse to decrease in amplitude by a particular percentage.

4. Discharge Energy: - Integrating the discharge pulse across time to determine the energy associated with each discharge event.

5. Statistical Analysis:

-Histogram Analysis: Making a histogram to see how the magnitudes or frequencies of the discharge are distributed

-Probability Density Function (PDF): Estimating the possibility that a discharge will occur within a range of possible magnitudes

STANDARTS

Guidelines and specifications for performing partial discharge measurements in high-voltage equipment are provided by the IEC 60270 standard. It includes test methodologies, measurement methods, calibration of instruments, data processing, and result interpretation. This standard's adherence guarantees uniform and precise testing procedures for determining insulation condition.

ADVANTAGES OF THE PARTİAL DİSCHARGE SYSTEM

- Maintenance Planning

Test results help determine equipment maintenance schedule. If PD activity is high, unexpected failures can be prevented by performing scheduled maintenance or repair work. Extending Equipment Life: Early detection prevents potential failures and extends equipment life. These advantages make segmented discharge tests an important tool for increasing the reliability of high-voltage equipment and reducing operating costs. PD tests are widely used in many areas such as the energy sector, industrial plants, railway systems and other high voltage applications.

APPLİCATİON AREAS OF THE PARTİAL DİSCHARGE SYSTEM

Power Transformers: PD tests are performed to evaluate the condition of the insulation systems of power transformers, identify potential faults or weaknesses, and ensure their reliable operation.

High-Voltage Cables: PD tests are performed to detect insulation defects of cables, evaluate their condition and identify possible cable failures before they occur.

Gas-Insulated Switchgear (GIS): PD tests help evaluate the insulation integrity of GIS equipment, detect insulation defects, and ensure that these systems operate safely and efficiently.

Rotating Machines: These tests are helpful in detecting insulation deterioration, monitoring the condition of the stator windings, and evaluatin the overall health of the machines

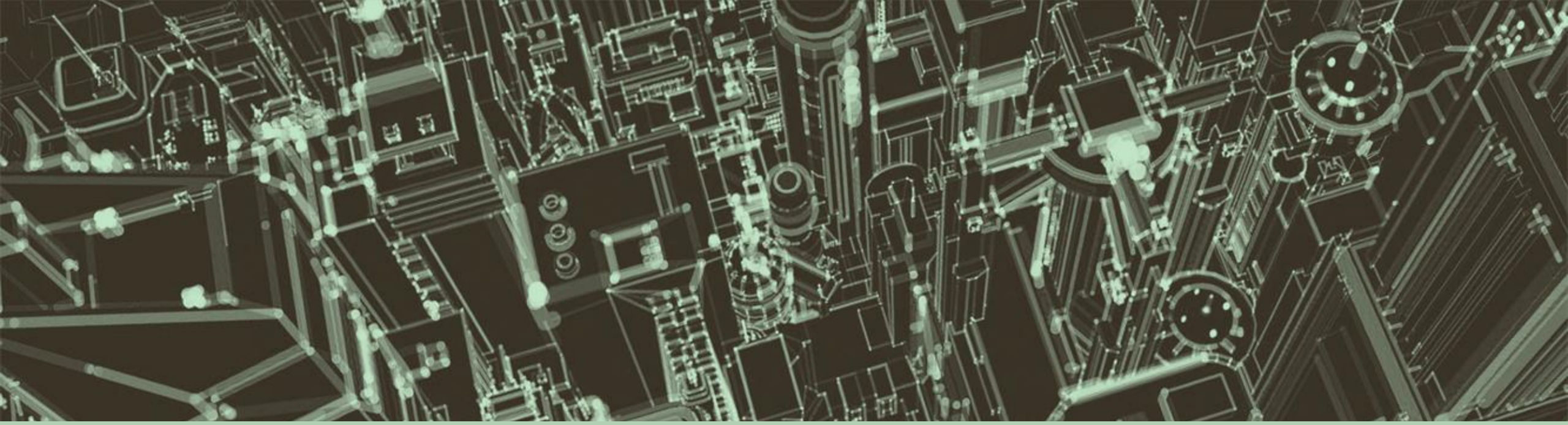
APPLİCATİON AREAS OF THE PARTİAL DİSCHARGE SYSTEM

- These are just a few examples of the application areas where the partial discharge system is utilized. PD testing is widely applied in the power industry, manufacturing plants, utilities, and other sectors to ensure the reliability, safety, and performance of high-voltage electrical equipment

CONCLUSION

- The results of a partial discharge test may provide important information about the state of the tested equipment. These conclusions comprise:

- Detection of Partial Discharge Activity
- Quantification of Discharge Magnitude
- Frequency of Discharge Occurrence:
- Location of Discharge Sources
- Comparative Analysis



PEHLA LABORATORY

Member of Short-Circuit Testing Liaison (STL)



PEHLA Laboratory

Member of Short-Circuit Testing Liaison (STL)



In March 1960, PEHLA was established as a civil law partnership. The laboratory is still active.

Pehla laboratory is an industrial type laboratory

The PEHLA Testing Division serves as an umbrella organization for high-performance testing laboratories in Germany and Switzerland.

- The PEHLA Laboratory, which stands for Partialentladungs-Hochleistungslabor (High-Performance Partial Discharge Laboratory), is a specialized facility dedicated to conducting advanced partial discharge testing. It serves as a center for research, analysis, and evaluation of electrical equipment in relation to partial discharge phenomena.
- The companies payments to its partners' laboratories to perform PEHLA tests.
- Separate accreditations and regular monitoring audits ensure privacy and information security.
- PEHLA test facilities provide development and type testing for domestic and international power system suppliers. All laboratories have accreditation (ISO/IEC 17025). Which is the standard that must be accredited to be considered technically authorized.
- Customers have access to 9 accredited PEHLA laboratories specializing in medium and high voltage for performing PEHLA tests.



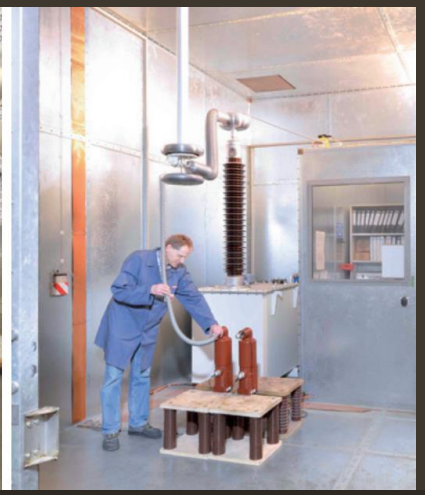


Pehla tests in 5 main areas

HIGH POWER TESTS

- Short circuit making and breaking tests
- Internal arc tests
- Capacitive current switching / Shunt reactor switching tests
- Load switching tests
- Synthetic tests
- Electrical endurance tests





High-power lab

The high-power testing laboratory features a 2800 MVA short-circuit test generator and oil-insulated power transformers. It enables making and breaking tests at various voltage and short-circuit current levels. A specialized dry-type power transformer facilitates peak-withstand current and short-time withstand current tests, reaching up to 250 kA and 100 kA r.m.s for three seconds. The facility includes an arcing test bay for conducting internal arcing tests on switchgear, containers, and substations. Capacitive tests, load current switching tests, and a wide range of other tests can be performed using different equipment such as reactors, resistors, and measurement tools.

HIGH VOLTAGE TESTS

- ❑ Impulse voltage (lightning/switching)
- ❑ Power frequency voltage
- ❑ Partial discharge
- ❑ DC voltage
- ❑ dielectric tests dry & wet
- ❑ Dielectric tests in climatic chambre
- ❑ Dielectric ageing



High-voltage testing laboratory

With the test facilities in our high voltage testing laboratory all dielectric and partial discharge tests for medium voltage equipment can be carried out. For sensitive partial discharge tests a special test chamber is available with a background level < 1 pC.

In order to offer on-site testing the high-voltage laboratory has mobile test equipment.

The high-voltage testing laboratory performs the following tests:

- Standard lightning impulse voltage tests
- Up to 800 kV
- Power-frequency voltage tests (Stationary up to 260 kV - Mobile up to 230 kV)
- Partial discharge tests (Stationary up to 150 kV - Mobile up to 230 kV)
- Degree of protection tests
- Tests on auxiliary and control circuits

HIGH CURRENT TESTS

- Temperature-rise tests

MECHANICAL TESTS

- Tightness tests
- Sound level tests
- Mechanical protection tests (IP and IK)
- Pressure tests
- Test of Power kinematic chain and position indicating kinematic chain

CLIMATIC TESTS

- Temperature limit tests / high and low temperature test
- Tests on switchgear to be used in severe climatic conditions
- Humidity test
- Insulator test





Temperature-rise lab

Tests up to 5000 A on switchgear. Automated and computer-controlled tests for precision and efficiency. Continuous monitoring of currents and temperatures. Options for up to 180 measuring points. Additional services include magnetic field measurement and thermal imaging.

Mechanical lab

Tests on medium and low voltage equipment. Functional tests (endurance, kinematic chain, interlocking). Measurement equipment for detailed data collection. Gas-tightness and pressure withstand tests for gas-filled equipment. Environmental tests (temperature, humidity, vibrations). Material testing (load tests, high-speed video recording).

Material lab

Climatic tests (-70°C to +150°C) on materials, components, and medium voltage switchgear panels. Temperature, humidity, and corrosion tests.

Overview of standards

- High-voltage switchgear and controlgear IEC 62271-1 IEC 62271-100 IEC 62271-102 IEC 62271-103 IEC 62271-104 IEC 62271-105 IEC 62271-106 IEC 62271-110 IEC 62271-111 IEC 62271-200 IEC 62271-201 IEC 62271-202 IEC 62271-203 IEC 62271-304 IEC 60529
- High-voltage test techniques IEC 60060-1 IEC 60060-2 IEC 60270
- Power transformers IEC 60076-5 IEC 60076-11
- High-voltage fuses IEC 60282-1 IEC 60282-2
- Bushings IEC 60137
- Insulators IEC 60660
- Instrument transformers IEC 61869-1 IEC 61869-2 IEC 61869-3
- Live working IEC 60832-1 IEC 60832-2 IEC 61230
- Low-voltage switchgear and controlgear IEC 60947-1 IEC 60947-2 IEC 60947-3
- ANSI / IEEE IEEE C37.04 ANSI C37.06 IEEE C37.09 ANSI C37.54 IEEE C37.60

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- Study on different techniques of partial discharge(PD) detection in power transformers winding: Simulation between paper and EPOXY resin using UHF method--International Journal of Conceptions on Electrical and Electronics Engineering--Irfan Ali Soomro and Md Nor Ramdon
- STUDY ON PULSE CURRENT OF TYPICAL PD MODELS IN POWER TRANSFORMER-- Guoli Wang, Yanpeng Hao, Yanming Li
- EXPERIENCE WITH OPTICAL PARTIAL DISCHARGE DETECTION--Michael Muhr, Robert Schwarz
- Fiber Optic Sensor for Acoustic Detection of Partial Discharges in Oil-Paper Insulated Electrical Systems--Julio Posada-Roman, Jose A. Garcia-Souto and Jesus Rubio-Serrano